

TECHNICAL REPORT: SOLAR-POWERED ELEVATOR SYSTEM

With Integrated Battery Energy Storage System (BESS) & Emergency Rescue System (ERS)

1. Project Overview & Consumption Profile

This project focuses on the decarbonization, energy efficiency, and resiliency of a building's high-consumption mechanical asset: the elevator system. The design transforms a standard elevator into an isolated, green Microgrid infrastructure capable of continuous operation and critical safety dispatch during utility grid blackouts.

The comprehensive energy matrix and load characterization are established as follows:

- Elevator Traction Motor Power Consumption: **18 kW**
- System Control & Ancillary Accessories Consumption: **2 kW**
- Total Peak Power Demand (P_{peak}): **20 kW**
- Daily Operational Window (Solar Peak Coefficient): **4 Hours**

Based on the operational window, the daily, monthly, and annual energy demand is calculated dynamically:

- Daily Energy Consumption: $E_{daily} = 20\text{ kW} \times 4\text{ Hours} = 80\text{ kWh/day}$
- Monthly Energy Consumption: $E_{monthly} = 80\text{ kWh} \times 30\text{ Days} = 2,400\text{ kWh}$
- Annual Energy Consumption: $E_{annual} = 2,400\text{ kWh} \times 12\text{ Months} = 28,800\text{ kWh}$

2. Photovoltaic (PV) Array Design & Sizing

To ensure rigorous industrial reliability and resource availability, the PV system is sized to offset the primary daily demand, utilizing high-efficiency monocrystalline solar modules.

- Solar Panel Output Rating: **390 W**
- Target Energy Generation Requirement: **20,000 Wh** (for the designated 4-hour operational peak sun hours)

The theoretical minimum and optimized configuration requirements are determined through the following mathematical criteria:

- Ideal Panel Count Calculation: $20,000\text{ Wh} / 390\text{ W} = 51.28\text{ panels}$
- Adjusting for an optimized configuration providing a 90% Base Load Offset to match localized physical constraints, a total of **46 pieces of 390W panels** are selected, yielding an optimized total peak capacity of **17.94 kW**.

String & Inverter Configuration

To mitigate localized shading losses, structural mismatch coefficients, and maximize the operational tracking efficiency of the 15000TL Inverter, the 46 panels are arranged into two primary asymmetrical sub-arrays utilizing 2 independent Maximum Power Point Tracking (MPPT) inputs:

Inverter MPPT Input	String Designation	Panels per String	Electrical Arrangement	Total Sub-Array Panels
MPPT Input I	String 1	12 Panels	Connected in Series	24 Panels (9.36 kW)
	String 2	12 Panels		
	String 3	11 Panels		
MPPT Input II	String 4	11 Panels	Connected in Series	22 Panels (8.58 kW)

3. Electrical Specifications & Cable Sizing

All component topologies, electrical nodes, and conductor cross-sections are engineered to comply strictly with the National Electrical Code (NEC) thermal and voltage drop regulations.

- Maximum Power Current (I_{mp}): **9.49 A**
- Short-Circuit Current (I_{sc}): **10.12 A**
- DC Side Protection Topology: Fast-acting melting overcurrent fuses rated proportionally to the panel short-circuit current to eliminate fault propagation risks.
- DC Cable Cross-Section: **4 mm²** copper conductors (optimized to ensure a cross-string voltage drop of $\Delta V \leq 1\%$).
- AC Side Main Interface: **220 V** / 3-Phase balanced system connection.
- AC Cable Cross-Section: **5 × 6 mm²** copper armored cable, sized to safely handle the nominal output current of the inverter under maximal thermal ambient stress.

4. Advanced Energy Recovery & Blackout Safety

The structural core innovation of this system lies in its integrated dual-layer energy storage and emergency rescue architecture, establishing robust localized security.

A. Regenerative Braking via Energy Saving Unit (ESU)

The traction framework utilizes an **18.5 kW Gearless Asynchronous AC Induction Motor** managed dynamically by a high-performance Variable Voltage Variable Frequency (3VF) Drive.

When the elevator cage travels downstairs with a heavy load factor, or upstairs under an empty/light load condition, the induction machine operates within its negative slip region, acting as a generator. Instead of dissipating this kinetic energy as waste thermal energy across a typical dynamic braking resistor grid, the system channels the power into an intelligent Energy Saving Unit (ESU), which filters, regulates, and redirects this recovered power to continuously cycle the main battery bank storage reserves.

B. Emergency Rescue System (ERS) for Blackout Conditions

To permanently negate passenger entrapment risks during utility grid blackout conditions, a dedicated, decoupled Blackout Emergency Rescue Device is implemented outside the primary control loop.

Operational Dispatch Protocol:

1. **Battery Pack Array:** Comprises 4 to 6 deep-cycle 12V battery blocks connected to establish a distinct safety power bus.
2. **Smart Directional Logic:** Upon sensing a total utility grid voltage collapse, the localized blackout board immediately initializes a micro-directional diagnostic sequence, executing brief upward and downward test movements.
3. **Current Minimization Selection:** The onboard microcontroller senses and evaluates the instantaneous torque current draw for both directions. It automatically selects the path of least mechanical resistance (dependent on whether the counterweight or the instantaneous cage mass is heavier), driving the elevator safely to the nearest immediate threshold floor and actuating the door mechanism to release passengers.

5. Core Engineering Conclusions

1. **Contactless Modern Control:** The high-speed elevator control topology is successfully implemented utilizing contactless logic arrays and microcontrollers, matching modern industrial automated standards.
2. **Modular Reliability:** The control panel implements a strict modular design principle, completely isolating critical safety functions (like the ERS) from standard operational distribution modules to simplify maintenance routines.
3. **Simulink Verified:** The formal mathematical model of the variable-frequency asynchronous traction drive was fully compiled, simulated, and validated within the MATLAB/Simulink environment.
4. **Energy Recovery Efficiency:** The implementation of the online regenerative battery charging grid achieves an advanced energy recovery rate, supplying up to 200 starts under high variable dynamic profiles without drawing power from the primary utility network.

6. System Single-Line Diagram (ETAP 2021)

The electrical single-line diagram modeled and simulated in ETAP 2021 illustrating the complete bus architecture, PV arrays, MPPT inverters, BESS, ERS, and elevator motor drive loops:

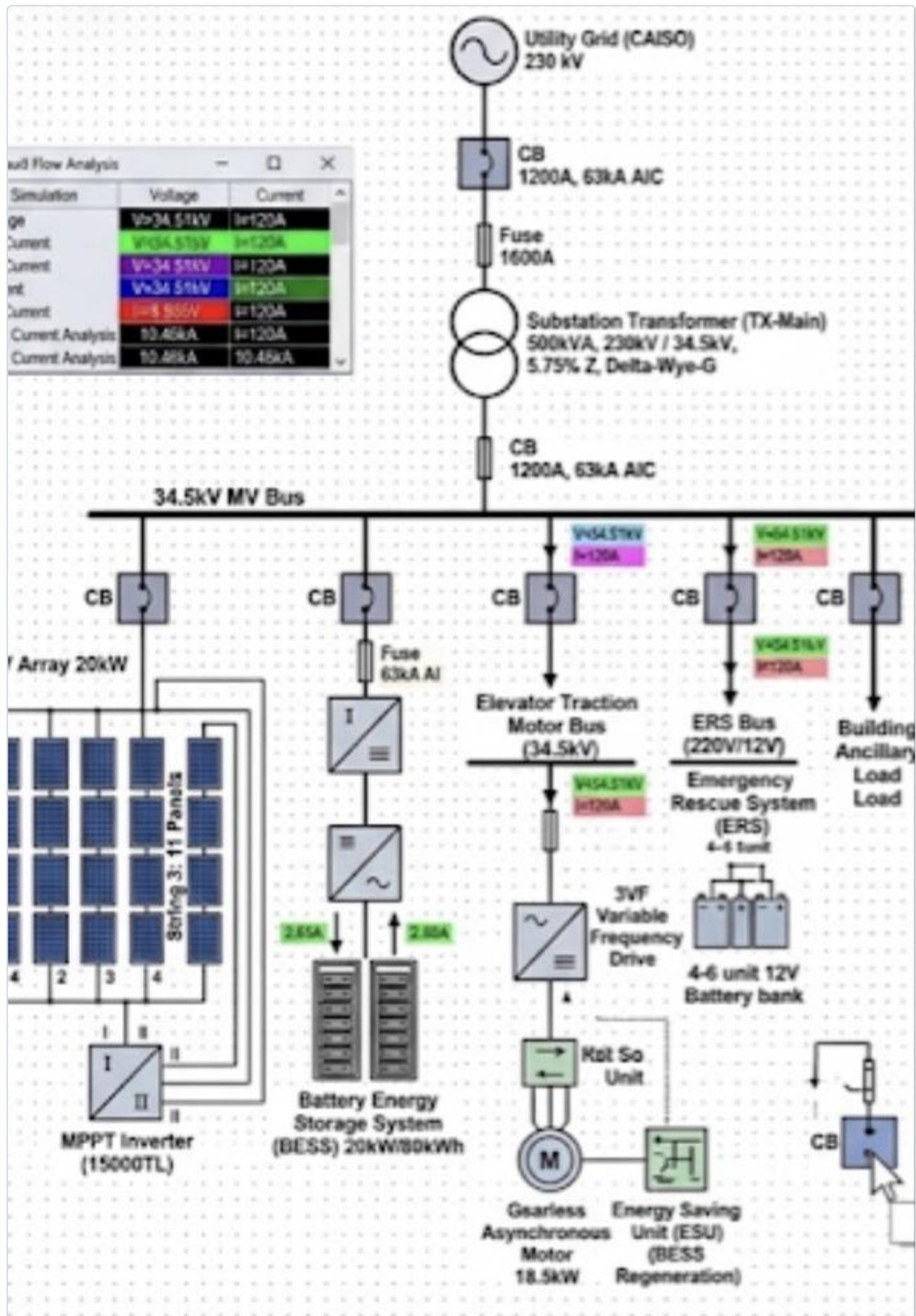


Figure 1: Complete Electrical Single-Line Diagram of the Solar-Powered Elevator Microgrid in ETAP 2021.